Reg. No. | | | | | | | | | | | [17MT43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

FUELS FURNACES AND REFRACTORIES

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least **TWO** questions from each part.
2. Missing data may be suitably assumed.
3. Draw neat sketches wherever necessary

PART – A

1.	**a.**	What is a fuel? Classify fuels. List the merits and limitations of fuels.	**08**
	b.	Define furnace. List the basic components of a furnace and the factors responsible for selection of furnaces.	**06**
2.	**a.**	What is coal? How coals are classified and briefly explain the coal preparation mechanisms.	**07**
	b.	With a neat sketch explain coal cleaning methods by using jigs.	**07**
3.	**a.**	Briefly explain about the various sources of liquid fuels.	**07**
	b.	Briefly explain the flue gas analysis using Orsat apparatus.	**07**
4.	**a.**	Describe the mechanism of combustion of pulverized fuel through burner.	**06**
	b.	A producer gas has the composition of H_2 = 16%, CH_4 = 1%, CO = 22%, CO_2 = 56%, O_2 = 1% and N_2 = 4%. Find out:	**08**
	i.	Minimum volume of air required for complete combustion of 100 liters of the gas;	
	ii.	Volumetric analysis of flue gas on dry basis and wet basis.	

PART - B

5.	**a.**	Briefly explain about the thermal shields and acoustic chambers in furnaces.	**06**
	b.	Explain the construction and working of a cupola furnace.	**08**
6.	**a.**	What is a refractory? List the properties of a refractory and briefly explain them.	**08**
	b.	Explain the manufacturing process, properties and applications of fire clay brick.	**06**
7.	**a.**	Briefly explain the various modes of heat transfer in a furnace.	**06**
	b.	Write about the source of heat loss in a furnace and explain the important waste heat recovery devices.	**08**
8.	**a.**	Briefly explain about furnace atmosphere.	**07**
	b.	Explain in detail about the pollution controlling devices.	**07**

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